OPEN

Localized Corneal Biomechanical Alteration Detected In Early Keratoconus Based on Corneal Deformation Using Artificial Intelligence

Xuan Chen, MD, Zuoping Tan, MD, PhD†, Yan Huo, MD*, Jiaxin Song, MD‡, Qiang Xu, MD†, Can Yang, MD†, Vishal Jhanji, MD, PhD§, Jing Li, MD, PhD||, Jie Hou, MD, PhD¶, Haohan Zou, MD, PhD‡, Gauhar Ali Khan, MD‡, Mohammad Alzogool, MD*, Riwei Wang, MD, PhD†,* and *Yan Wang, MD, PhD*‡#***

Purpose: This study aimed to develop a novel method to diagnose early keratoconus by detecting localized corneal biomechanical changes based on dynamic deformation videos using machine learning.

Design: Diagnostic research study.

Methods: We included 917 corneal videos from the Tianjin Eye Hospital (Tianjin, China) and Shanxi Eye Hospital (Xi'an, China) from February 6, 2015, to August 25, 2022. Scheimpflug technology was used to obtain dynamic deformation videos under forced puffs of air. Fourteen new pixel-level biomechanical parameters were calculated based on a spline curve equation fitting by 115,200-pixel points from the corneal contour extracted from videos to characterize localized biomechanics. An ensemble learning model was developed, external validation was performed, and the diagnostic performance was compared with that of existing clinical diagnostic indices. The performance of the developed machine learning model was evaluated using precision, recall, F1 score, and the area under the receiver operating characteristic curve.

Results: The ensemble learning model successfully diagnosed early keratoconus (area under the curve = 0.9997) with 95.73% precision, 95.61% recall, and 95.50% F1 score in the sample set (n = 802). External validation on an independent dataset (n = 115) achieved 91.38% precision, 92.11% recall, and 91.18% F1 score. Diagnostic accuracy was significantly better than that of existing clinical diagnostic indices (from 86.28% to 93.36%, all $P < 0.01$).

Conclusions: Localized corneal biomechanical changes detected using dynamic deformation videos combined with machine learning algorithms were useful for diagnosing early keratoconus. Focusing on localized biomechanical changes may guide ophthalmologists, aiding the timely diagnosis of early keratoconus and benefiting the patient's vision.

Key Words: dynamic deformation videos, early keratoconus, localized biomechanics, machine learning, pixel-level data

(*Asia Pac J Ophthalmol (Phila)* 2023;12:574–581)

Submitted August 1, 2023; accepted September 27, 2023.

From the *School of Medicine, Nankai University, Tianjin, China; †Wenzhou University of Technology, Wenzhou, Zhejiang, China; ‡Clinical College of Ophthalmology, Tianjin Medical University, Tianjin, China; §Department of Ophthalmology, University of Pittsburgh School of Medicine, Pittsburgh, PA; ||Shanxi Eye Hospital, Xi'an People's Hospital, Xi'an, China; ¶Jinan Mingshui Eye Hospital, Ji'nan, Shandong, China; #Tianjin Eye Hospital, Tianjin Key Lab of Ophthalmology and Visual Science, Tianjin Eye Institute, Nankai University Affiliated Eye Hospital, Tianjin, China; and **Nankai Eye Institute, Nankai University, Tianjin, China.

X.C., and Z.T. are co-first authors.

R.W. and Y.W. are the corresponding authors.

The data supporting this study's findings are available from the corresponding author upon reasonable request.

X.C. and Z.T. contributed equally to this work. X.C. and Z.T. had full access to the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: X.C., Z.T., V.J., R.W., and Y.W. Data acquisition or research execution: X.C., Z.T., Y.H., J.S., J.L., J.H., and H.Z. Data analysis or interpretation: X.C., Z.T., Y.H., Q.X., C.Y., V.J., H.Z., R.W., Y.W. Drafting of the manuscript: X.C., Z.T.. Critical revision of the manuscript for important intellectual content: X.C., Z.T., Y.H., J.S., V.J., G.A.K., M.A., R.W., and Y.W.

This work was supported by the National Program on Key Research Project of China (2022YFC2404502), the National Natural Science Foundation of China (82271118), the Tianjin Key Medical Discipline (Specialty) Construction Project (TJYXZDXK-016A), Nankai University Eye Institute (NKYKD202209), and Wenzhou University of Technology (KY202204).

The authors have no conflicts of interest to disclose.

Address correspondence and reprint requests to: Yan Wang, Tianjin Eye Hospital, Tianjin Key Lab of Ophthalmology and Visual Science, Tianjin Eye Institute, Nankai University Affiliated Eye Hospital, Tianjin, China; Nankai Eye Institute, Nankai University, Tianjin, China; Clinical College of Ophthalmology, Tianjin Medical University, Tianjin, China, No 4. Gansu Road, Heping District, Tianjin 300020, China. E-mail: wangyan7143@vip.sina.com; Riwei Wang, Wenzhou University of Technology, No. 1 Landscape Avenue, Chashan Street, Ouhai District, Wenzhou 325035, Zhejiang, China. E-mail: wangrw@wzu.edu.cn

Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, https://journals.lww.com/apjoo.

ISSN: 2162-0989

DOI: 10.1097/APO.0000000000000644

INTRODUCTION

Keratoconus (KC) is a corneal ectatic disorder characterized by progressive central or paracentral corneal thinning and protrusion, resulting in irregular astigmatism and vision loss.[1] Typically, KC is diagnosed upon clinical examination and corneal tomography. Despite advancements in corneal imaging techniques, diagnosing early-stage KC, including subclinical, suspected, or forme fruste KC, remains challenging because there are no obvious abnormalities in corneal morphology in the earliest stages.[2–4] Without timely detection at an early stage, the disease continues to progress. The progression of the condition can have severe consequences and often requires corneal transplantation.[5,6]

Studies suggest that corneal morphologic changes, such as corneal thinning and protrusion, can occur as a result of weakened corneal biomechanics.[7–9] Furthermore, previous studies have shown that the Corvis Biomechanical Index (CBI), Tomographic Biomechanical Index (TBI), and other composite biomechanical-related parameters are useful tools for the accurate diagnosis of KC.[10,11] However, such results have been inconsistent in diagnosing early KC. Vinciguerra et al reported biomechanical abnormalities in both eyes with unilateral KC,[12] whereas Steinberg et al reported no significant differences in the biomechanical parameters between subclinical KC and normal controls.[13] Thus, whether corneal biomechanical changes occur during the early stages of KC remains controversial. Previous studies have evaluated device-specific *in vivo* biomechanical parameters of the entire corneal information, which may be one reason for the underperformance of biomechanical indices in diagnosing early KC.[14–16]

Histopathological studies have shown that fragmentation and degeneration of collagen fibers resulting from proteoglycan degeneration could result in subtle biomechanical changes in localized KC.[17,18] Therefore, identifying local biomechanics may allow for the detection of variation before significant corneal topographic changes occur. However, existing *in vivo* biomechanical measurement devices using air-puff testing only reflect the deformation of the entire cornea rather than localized measurements. Pixel-level data is a good way to detect subtle changes in the image profiles owing to the larger number of data points available at the pixel level than can be calculated on a millimeter scale.

Analysis of pixel-level data has not been previously reported as a method of detecting corneal pathology, possibly because data access is limited by commercial devices.[19] Pixel-level data analysis can be accomplished using machine learning (ML) algorithms,[20,21] a branch of artificial intelligence (AI) that autonomously imitates human learning.[22] The concept of ML has previously been applied in ophthalmology to improve KC diagnostics.[23–26] Accordingly, this study aimed to use pixel-level data calculated from corneal dynamic deformation videos in conjunction with ML algorithms to introduce a new way of characterizing *in vivo* localized corneal biomechanical properties and develop a novel and intelligent method to diagnose early KC.

METHODS

Data Collection

Initial corneal dynamic deformation videos were collected using Corneal Visualization Scheimpflug Technology (Corvis ST; Oculus, Wetzlar, Germany)[27] with examination dates from February 6, 2015 to August 25, 2022. Corvis ST is an ultra-high-speed Scheimpflug camera with a capture speed of 4330 frames per second in an 8 mm-diameter zone along the horizontal meridian of the cornea in response to a puff of air, obtaining dynamic force deformation videos. Measurements were repeated 3 times for each eye, and the results with a "QS" (quality control) label of "OK" were selected for analysis. Videos with missing information, inaccessible videos, or videos with quality issues were excluded from this study. In addition to dynamic deformation videos, the following examinations and characteristics were recorded: age, sex, uncorrected distance visual acuity, corrected distance visual acuity (CDVA), objective and manifest refraction, noncontact intraocular pressure, corneal tomography examination, slit-lamp examination, and fundus examination.

Ethical approval and participant consent

The Ethics Committee of Tianjin Eye Hospital approved the study protocol (number: 2022021), which adhered to the principles of the Declaration of Helsinki. All participants provided informed consent for data use. This study followed the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis reporting guidelines for diagnostic studies.

Participant Inclusion and Grouping

A total of 917 eyes were recruited for the present study. Furthermore, 917 corneal videos were collected to calculate new pixel-level parameters to develop an ML model for the diagnosis of early KC. Participants of the same ethnicity from 2 clinics in 2 cities within the same country were selected to include variability from different regions: 802 corneal videos from the Tianjin Eye Hospital (sample database) and 115 corneal videos from the Shanxi Eye Hospital (validation database).

Two senior ophthalmologists (Y.W. and J.L.) marked and grouped each sample. Patients with bilateral KC were included in the KC group, as they met the following inclusion criteria: history of myopia and astigmatism, CDVA <20/20, at least 1 positive sign of KC on slit-lamp biomicroscopy (thinning of the corneal stroma, forward protrusion, Vogt striae, Fleischer ring, or Munson sign), and abnormal corneal tomography features including an inferior-superior asymmetry value of >3.0 diopters (D) on the anterior corneal surface within a 3 mm radius or intereye difference of >1.0 D in the anterior central curvature.[1] Participants in the early KC group had clinical features of KC in 1 eye, whereas the fellow eye had normal slitlamp, retinoscopy, and topographic evaluations (inferior-superior value of the anterior corneal surface within 3 mm radius <1.45 D); no asymmetry bowtie pattern; no focal or inferior steepening pattern; and CDVA of 20/20 or better. Fellow eyes with normal clinical characteristics were included in the study.[28] The third group of normal control participants (NL) had a CDVA of 20/20 or better and no

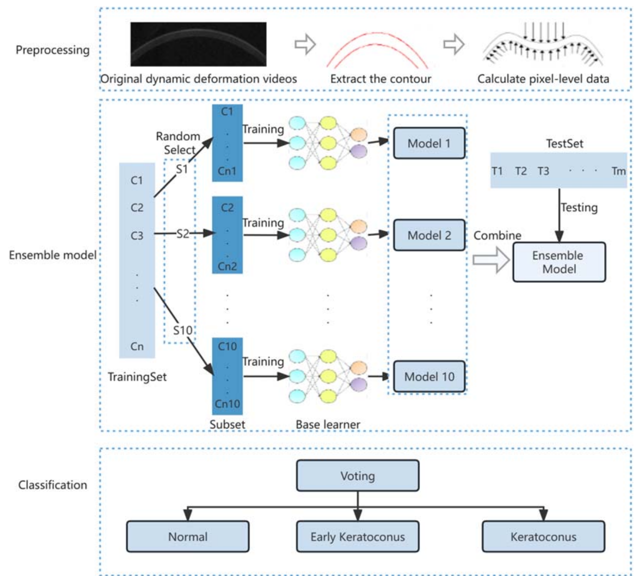

FIGURE 1. Flow chart of machine learning. Preprocessing, development of the ensemble learning model, and classification into keratoconus, early keratoconus, and normal eyes.

ocular abnormalities other than myopia. Before the examinations, participants were instructed not to wear soft or hard contact lenses for more than 2 or 4 weeks, respectively. No participant reported a history of ocular surgery, trauma, ocular diseases (except refractive error and primary KC), connective tissue diseases, or systemic diseases. Finally, 400 KC, 117 early KC, and 400 NL corneal videos were included. Of the samples, 687 were used for training and 115 for testing, with 115 independent samples for external validation.

Model construction, evaluation, and external validation

The ML flowchart is shown in Figure 1. Corneal contour pixels were extracted from dynamic deformation videos. The curve equation of the corneal contours was fitted using 115,200 (576 × 200) pixel points and subsequently calculated new parameters based on the spline theory and renamed them. The sample database was divided into training and testing sets, and 10-fold cross-validation was performed. In the training set, the dataset was sampled randomly, which yielded 10 mutually exclusive subsets. Each subset was trained with one base learner, and each base learner model was a 5-layer feed-forward neural network. These 10 base learners were combined as an ensemble learning model. In the testing set, the voting strategy was used to determine the sample category, and each sample category was voted for by 10 base learners. The one with the most votes was used as the final category for the testing sample. If both categories received the same number of votes, the sample was selected randomly.[29] The detailed process is illustrated in Figure 2. Considering each category and all others as dichotomous data models, the true-positive (TP) and false-positive (FP) rates were calculated, resulting in a visual receiver operating characteristic (ROC) curve. The area under the ROC curve (AUC), precision, recall, and F1 score were used to evaluate the performance of the ML model.

External validation was performed on an independent dataset from another city to evaluate the model's generalizability. The diagnostic accuracy of the model was compared with that of common clinical diagnostic indices, including Belin/Ambrósio deviation (BAD-D), CBI, and TBI.

Statistical Analysis

All statistical analyses were performed using the Statistical Package for the Social Sciences software version 26.0 (IBM Corp, Armonk, NY). Quantitative characteristics are expressed as the minimum and maximum values, mean, and SEM. The Kolmogorov–Smirnov test was applied to check data distribution. Analysis of variance was used to test differences in parameters among the datasets. Statistical significance was set at $P < 0.05$. The ROC curve was used to analyze the ability of the BAD-D, CBI, and TBI to diagnose early KC and calculate the cutoff value for the best diagnostic ability.

Precision represents the proportion of correctly predicted quantities in the predicted classification. Recall represents the proportion of correctly predicted quantities in the true classification. The F1 score represents the harmonic mean of the precision and recall. Accuracy, sensitivity, specificity, and precision were calculated to evaluate the performance of the BAD-D, CBI, and TBI. The formulas for precision, recall, F1 score, accuracy, sensitivity, specificity, and precision are provided in Supplementary Digital Content File 1, http://links.lww.com/APJO/A295.

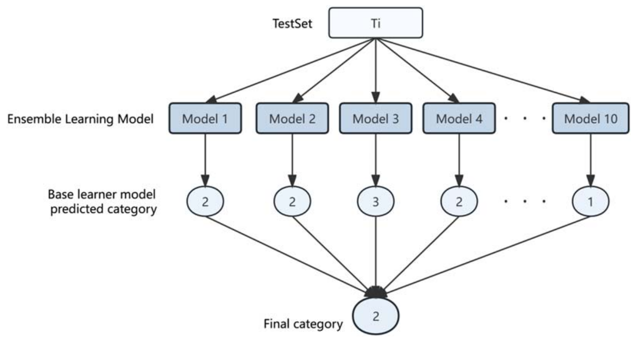

FIGURE 2. The voting process in detail.

RESULTS

We included 917 eyes in this study. The sample database included 350 NL, 102 early KC, and 350 KC eyes for training and testing; the validation database included 50 NL, 15 early KC, and 50 KC eyes for external validation. The mean ages of patients with normal eyes were 24.30 ± 6.55 y and 23.45 ± 5.84 y in the sample and validation database, respectively. The mean ages of early KC patients were 23.25 ± 5.70 y and 23.37 ± 6.80 y in the sample and validation database, respectively, and those of KC patients were 23.24 ± 5.84 y and 21.96 ± 6.51 y in the sample and validation database, respectively. Table 1 summarizes and analyzes the participants' baseline data.

Fourteen new pixel-level parameters extracted from corneal dynamic deformation videos characterizing biomechanics were described with good repeatability. These parameters were calculated and named based on 115,200 (576 × 200) corneal contour pixel points, including variations in full contour length at the first and second applanations, time at the first and second applanations, depth of the thinnest point at the first applanation, second applanation, and maximum depression, length at the first and second applanation, relative displacement of the thinnest point (1 mm and 2 mm), peak distance, maximum curvature, and maximum depression area. The definitions, calculation methods, and values are listed in Table 2.

We developed an ensemble learning model to classify based on localized biomechanics. The visualized ROC for the classification model is shown in Figure 3; the AUCs were all nearly 1.0000 (KC = 0.9997, early KC = 1.0000, and NL = 0.9997). The ensemble learning model successfully diagnosed early KC with 95.73% precision, 95.61% recall, and 95.58% F1 score for the sample dataset and with 91.38% precision, 92.11% recall, and 91.18% F1 score for the external validation dataset. Confusion matrices are shown in Figure 4.

As shown in Table 3, the accuracy of the ensemble learning model (95.73%) for the diagnosis of early KC was significantly better than that of the BAD-D (93.36%, $P < 0.01$), CBI (88.72%, $P < 0.01$), and TBI (86.28%, $P < 0.01$).

DISCUSSION

In this study, we described more localized and detailed corneal biomechanical characteristics using dynamic deformation videos combined with AI. By detecting localized

TABLE 1. Participants' Baseline Characteristics

	Sample dataset			Validation dataset			***P***
Characteristics	NL	early KC	KC	NL	early KC	KC	–
Sex (M/F)	208/142	73/29	200/150	26/24	12/3	36/14	–
Age	24.30 ± 6.55	23.25 ± 5.70	23.24 ± 5.84	23.45 ± 5.84	23.37 ± 6.80	21.96 ± 6.51	P^a = 0.387 P^b = 0.939 P^c = 0.158
K1 (D)	42.18 ± 1.38	42.50 ± 1.35	46.41 ± 4.28	42.36 ± 1.20	42.85 ± 1.57	46.66 ± 4.61	P^a = 0.401 P^b = 0.348 P^c = 0.704
K2 (D)	43.44 ± 1.44	43.87 ± 1.54	50.02 ± 5.03	43.53 ± 1.40	44.07 ± 1.93	50.45 ± 5.00	P^a = 0.679 P^b = 0.653 P^c = 0.573
Kmax (D)	43.95 ± 1.47	44.84 ± 1.80	56.53 ± 7.58	44.02 ± 0.44	45.48 ± 1.95	57.40 ± 7.70	P^a = 0.773 P^b = 0.208 P^c = 0.455
CCT	557.53 ± 23.87	514.57 ± 32.03	470.91 ± 36.34	557.14 ± 32.33	511.27 ± 41.75	472.49 ± 37.72	P^a = 0.918 P^b = 0.721 P^c = 0.778
bIOP	16.19 ± 2.20	16.00 ± 1.84	15.59 ± 1.75	15.32 ± 2.09	14.73 ± 2.05	14.24 ± 1.89	P^a = 0.560 P^b = 0.592 P^c = 0.109
KI	1.02 ± 0.02	1.04 ± 0.03	1.22 ± 0.12	1.03 ± 0.02	1.04 ± 0.04	1.21 ± 0.12	P^a = 0.438 P^b = 0.964 P^c = 0.621
CKI	1.01 ± 0.01	1.01 ± 0.01	1.08 ± 0.06	1.01 ± 0.01	1.01 ± 0.01	1.08 ± 0.06	P^a = 0.511 P^b = 0.629 P^c = 0.676
BAD-D	0.75 ± 0.48	2.17 ± 0.99	8.97 ± 4.21	0.81 ± 0.39	2.42 ± 1.18	8.89 ± 3.90	P^a = 0.444 P^b = 0.371 P^c = 0.909
CBI	0.17 ± 0.16	0.59 ± 0.28	0.96 ± 0.10	0.14 ± 0.12	0.67 ± 0.35	0.97 ± 0.05	P^a = 0.203 P^b = 0.305 P^c = 0.381
TBI	0.12 ± 0.15	0.63 ± 0.39	0.99 ± 0.02	0.14 ± 0.16	0.63 ± 0.36	1.00 ± 0.00	P^a = 0.372 P^b = 0.974 P^c = 0.447

The difference between the sample and validation datasets were analyzed using analysis of variance (P^a: NL; P^b: early KC; P^c: KC).

BAD-D indicates Belin/Ambrósio deviation; bIOP, biomechanically intraocular pressure; CBI, Corvis Biomechanical Index; CCT, center corneal thickness; CKI, center keratoconus index; D, diopters; F, female; K1, flat keratometry of the corneal anterior surface; K2, steep keratometry of the corneal anterior surface; KC, keratoconus; KI, keratoconus index; Kmax, maximum keratometry; M, male; NL, normal; TBI, Tomographic and Biomechanical Index.

TABLE 2. Definition, Calculation Methods, and Value of New Pixel-Level Biomechanical Parameters

Parameters	Definition	Calculation
Variation of full contour length at first applanation [Px]	The variation of corneal anterior surface contour length (original acquisition range of the instrument) between the first applanation and the initial state	The integration of all pixel points of the first frame minus the integration of all pixel points of the first applanation frame
Variation of full contour length at second applanation [Px]	The variation of corneal anterior surface contour length (original acquisition range of the instrument) between the second applanation and the initial state	The integration of all pixel points of the first frame minus the integration of all pixel points of the second applanation frame
Time at first applanation [ms]	Time to reach the first applanation	The number of first applanation frames multiplied by the (total video time/total video frame count)
Time at second applanation [ms]	Time to reach the second applanation	The number of second applanation frames multiplied by the (total video time/total video frame count)
Depth of the thinnest point at first applanation [Px]	The variation of the thinnest point vertical displacement between the first applanation and the initial state	The y-value of the thinnest point of the cornea in the first applanation frame minus the y-value of the thinnest point of the cornea in the first frame
Depth of the thinnest point at second applanation [Px]	The variation of the thinnest point vertical displacement between the second applanation and the initial state	The y-value of the thinnest point of the cornea in the second applanation frame minus the y-value of the thinnest point of the cornea in the first frame
Depth of the thinnest point at maximum depression [Px]	The variation of the thinnest point vertical displacement between the maximum depression and the initial state	The y-value of the thinnest point of the cornea in the maximum depression frame minus the y-value of the thinnest point of the cornea in the first frame
Length at first applanation [Px]	Segment length at first applanation	All pixel points on the applanation segment of the first applanation frame
Length at second applanation [Px]	Segment length at second applanation	All pixel points on the applanation segment of the second applanation frame
Relative displacement of the thinnest point [1 mm]	The ratio of the variation of the thinnest point vertical displacement at maximum depression to the mean variation of the thinnest point within 1 mm vertical displacement	Vertical displacement of the thinnest point at the maximum depression frame/mean vertical displacement at each 72-pixel point from the left and right of the thinnest point, 576 pixels/8 mm = 72 pixels/mm
Relative displacement of the thinnest point [2 mm]	The ratio of the variation of the thinnest point vertical displacement at maximum depression to the mean variation of the thinnest point within 2 mm vertical displacement	Vertical displacement of the thinnest point at the maximum depression frame/mean vertical displacement at each 144-pixel point from the left and right of the thinnest point, 576 pixels/8 mm = 72 pixels/mm
Peak distances [Px]	The horizontal distance of the highest point on either side of the thinnest point at maximum depression	The x-value corresponding to the maximum y-value to the right of the thinnest point at the maximum depression frame minus the x-value corresponding to the maximum y-value to the left of the thinnest point at the maximum depression frame
Maximum curvature [Px^{-1}]	The maximum value of the curvature of the thinnest point during the whole dynamic deformation process	Calculate the curvature of the thinnest point for each frame of 139 frames separately and take the maximum value
Maximum depression area [Px^2]	Area of deformation from maximum depression	Area enclosed by the corneal anterior surface profile curve of the first frame and the corneal anterior surface profile curve of the maximum depression frame

KC indicates keratoconus; NL, normal; Px, pixel.

biomechanical alternations, particularly in early KC, we provided a novel way to resolve the existing controversy on whether corneal biomechanical changes occur at an early stage of KC.[12–16] More importantly, we developed an ML model and diagnosed early KC with 95.73% precision, 95.61% recall, and 95.58% F1 score in the sample set, which is superior to the current and widely accepted approaches,[2,30] thereby improving the accuracy of diagnosis and reducing the risk of misdiagnosis.

Our findings supported the hypothesis that biomechanical changes in early KC are localized rather than global. Dynamic corneal deformation videos contained direct information from the cornea and avoided the measurement error caused by data conversion to the maximum extent. According to the theory that biomechanical weakness can produce greater deformation, our pixel-level biomechanical parameters were obtained based on spline curve equation fitting by real corneal contours extracted after segmenting dynamic deformation videos. Interestingly, our study detected more detailed changes in biomechanics by calculating the parameters based on 72-pixel points within 1 mm (576 px/8 mm = 72 px/mm), which showed high reliability in distinguishing between normal and early KC. This could reflect more local changes in biomechanics, which differ from the analysis of whole corneal biomechanical information. Moreover, good performance of independent external validation with 91.38% precision, 92.11% recall, and 91.18% F1 score in diagnosing early KC ensures the generalizability and stability of our findings; that is, detecting localized biomechanical changes is useful for diagnosing early KC. The earlier the diagnosis of KC is made, the more the patient's vision will benefit, thereby preventing serious consequences, such as blindness, that could affect the quality of life.

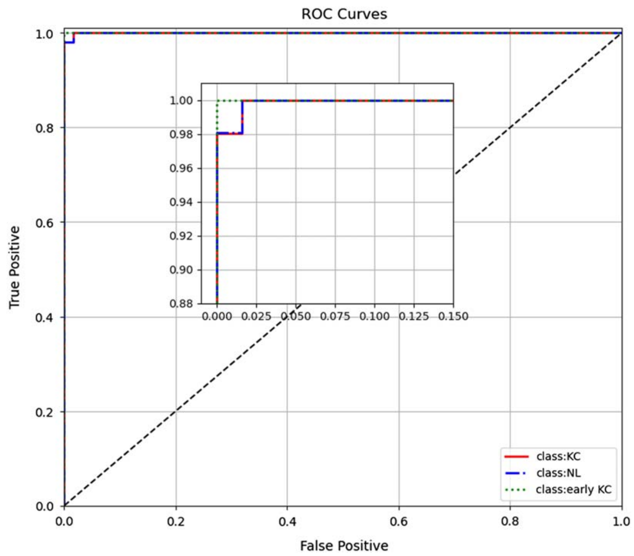

FIGURE 3. Visualization of the receiver operating characteristic curve. A dichotomous data model was used to determine the TPR and the FPR. FPR and TPR are represented on the x-axis and y-axis for a visualized receiver operating characteristic curve. FPR indicates false-positive rate; KC, keratoconus; NL, normal; ROC, receiver operating characteristic ; TPR, true-positive rate.

However, the existing device-specific parameters reflect the biomechanical properties of the entire cornea,[31] providing a more generalized picture; they cause difficulty in sensitively detecting localized biomechanical changes produced by focal areas of biomechanical weakness. A typical example is the Corvis ST, which reflects the biomechanical characteristics of the entire cornea *in vivo.* The CBI is based on dynamic corneal response parameters and Ambrósio Relational Thickness to the horizontal profile. This index has been confirmed to detect typical KC accurately; however, it is not sufficiently reliable for early disease stages.[32] We introduced 14 new pixel-level biomechanical parameters associated with localized corneal biomechanics and innovatively applied this technology for early KC detection. Our findings indicated that localized corneal biomechanical changes detected in pretomographic KC may lead to the development of clinically measurable *in vivo* parameters.

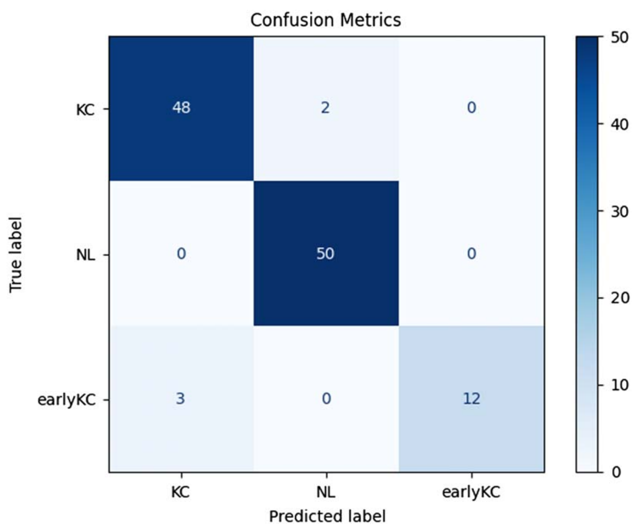

FIGURE 4. Confusion matrices of machine learning model. KC indicates keratoconus; NL, normal.

Our study demonstrated that evaluating localized corneal biomechanics has an advantage in detecting early KC compared with obtaining the existing clinical diagnostic indices, in which BAD-D had accuracy, sensitivity, and specificity levels of 93.36%, 71.3%, and 99.7%, respectively, with a cutoff value of 1.61. When the cutoff value of CBI was 0.4295, the accuracy, sensitivity, and specificity levels for the diagnostic ability of early KC were of 88.72%, 70.3%, and 94.0%, respectively; when the cutoff value for TBI was 0.2895, the accuracy, sensitivity, and specificity levels for diagnosing early KC were 86.28%, 75.2%, and 89.4%, respectively. Meanwhile, considering that the posterior corneal elevation at the thinnest point (PTE) is also sensitive to diagnosing KC,[33,34] the ability of the PTE to diagnose early KC was analyzed, and we found that the AUC was 0.578. When the cut-off value of PTE was 26.5, the accuracy, sensitivity, and specificity levels for the early KC diagnosis were 50.79%, 96.5%, and 42.3%, respectively. These indicators have been widely used to distinguish classic KC, and the diagnosis is relatively easily performed,[35] probably owing to significant changes in classic KC morphology. However, they are not as useful for early KC (Table 3). In the present study, localized corneal biomechanical changes were successfully detected, and early KC was distinguished solely based on corneal biomechanics, independent of corneal tomography. Notably, the results do not suggest that corneal tomography examination could be ignored; the critical role of corneal biomechanics in early KC identification is emphasized.

We used an AI-based learning algorithm for localized biomechanical assessment. Although the use of AI in KC detection is increasing, performing external validation generally decreases diagnostic accuracy due to factors such as the generalization of the model and regional differences in keratoconus features.[24] Most studies to date only adopted a certain type of classifier, inevitably introducing classification bias.[16,26] Our study carefully avoided such issues. The ensemble learning model simultaneously evaluated 10 classifiers on the same dataset and obtained the final classification results based on voting, helping avoid selection bias and improving accuracy. Ultimately, our study achieved excellent classifications (AUC nearly 1.000) of KC, early KC, and NL using AI. Performing external validation is essential, and our model's diagnostic accuracy remained high after external validation with an independent dataset from another city, demonstrating its generalizability. Although its clinical application will need to be validated in larger populations, the development of this novel diagnostic model will help automatically detect KC at an early stage and accelerate the diagnostic process for early KC.

Unlike the approach used in other studies,[24] the ML model used in our study is a 3-classification model to prevent grouping bias. Previous studies have focused on changes in biomechanics in early KC using pairwise groupings and reported the correlation between corneal parameters and KC.[12,14] The weak correlation reported by these studies could not accurately determine whether an abnormality was real or a measurement error, making diagnosing early KC more difficult. Tan et al have developed a novel method for diagnosing KC using AI by implementing only dynamic corneal deformation videos to learn the biomechanics of KC.[36]

TABLE 3. Diagnostic Accuracy of BAD-D, CBI, and TBI and Comparisons With ML Model

	Early KC						KC					
	AUC	Cut-off	Accuracy, %	Sensitivity, %	Specificity, %	*P*	AUC	Cut-off	Accuracy, %	Sensitivity, %	Specificity, %	*P*
BAD-D	0.892	1.61	93.36	71.30	99.70	< 0.01	0.999	3.655	99.49	98.94	98.06	< 0.01
CBI	0.876	0.4295	88.72	70.30	94.00	< 0.01	0.981	0.8595	96.53	93.67	95.15	< 0.01
TBI	0.864	0.2895	86.28	75.20	89.40	< 0.01	0.954	0.8915	99.87	99.74	89.32	< 0.01

AUC indicates area under the receiver operating characteristic curve; BAD-D, Belin/Ambrósio Deviation; CBI, Corvis Biomechanical Index; KC, keratoconus; ML, machine learning; TBI, Tomographic and Biomechanical Index.

However, the present study introduces novel pixel-level parameters to analyze the localized corneal biomechanics and further extract the biomechanical features of early KC. As KC is generally considered a bilateral and asymmetric disease,[6] the fellow eye with relatively normal tomography in unilateral KC serves as a model for studying early KC. Terms such as subclinical KC, forme fruste KC, preclinical KC, and KC suspect sometimes overlap.[2–4] We used "early KC" to maintain consistency because the inclusion criteria for the group of eyes in the present study are consistent with those of Reddy JC et al[28] The results of the present study indicate that detecting subtle biomechanical changes before the development of corneal ectasia has significant implications for recognizing early KC, providing a more practical way for KC screening before corneal refractive surgery. The changes in biomechanical properties have been well reported not only in KC but also in various corneal diseases. This highlights the importance of corneal biomechanics during disease initiation and progression.[9]

Our study also had some limitations. First, it is difficult to determine whether one ML technique offers significant advantages over another; therefore, we advocate sharing datasets for comparison. Further, accurate assessment of biomechanical properties *in vivo* is challenging because of the complexity of viscoelastic biomechanical behavior and the influence of confounding factors such as central corneal thickness and intraocular pressure. Future studies are needed to combine fundamental research findings by exploring the actual biomechanical properties. This was a cross-sectional study, and our group is conducting longitudinal analyses to explore the relationship between the biomechanical properties and different severities of KC to predict disease progression. Meanwhile, we plan to combine important indexes of corneal topographic features to improve the clinical value.

In conclusion, pixel-level parameters calculated from dynamic deformation videos appear to be useful in providing local corneal information and detecting localized biomechanical changes to identify early KC intelligently. The ML model showed promising results, achieving an accurate diagnosis of early KC with precision, recall, and F1 score above 95%, which was better than other clinical diagnostic indicators. Independent external validation also showed high accuracy, demonstrating the model's clinical generalizability. This new method can provide guidance to ophthalmologists in the timely diagnosis of early KC, thereby benefiting the patient's vision.

REFERENCES

1. Rabinowitz YS. Keratoconus. *Surv Ophthalmol*. 1998;42:297–319.
2. Santodomingo-Rubido J, Carracedo G, Suzaki A, et al. Keratoconus: an updated review. *Cont Lens Anterior Eye*. 2022;45:101559.
3. Klyce SD. Chasing the suspect: keratoconus. *Br J Ophthalmol*. 2009;93: 845–847.
4. Henriquez MA, Hadid M, Izquierdo L Jr. A systematic review of subclinical keratoconus and forme fruste keratoconus. *J Refract Surg*. 2020;36:270–279.
5. Wisse RPL, Simons RWP, Van Der Vossen MJB, et al. Clinical evaluation and validation of the Dutch Crosslinking for keratoconus Score. *JAMA Ophthalmol*. 2019;137:610–616.

6. Gomes JAP, Tan D, Rapuano CJ, et al. Global consensus on keratoconus and ectatic diseases. *Cornea*. 2015;34:359–369.
7. Meek KM, Tuft SJ, Huang Y, et al. Changes in collagen orientation and distribution in keratoconus corneas. *Invest Ophthalmol Vis Sci*. 2005;46: 1948–1956.
8. Piñero DP, Alio JL, Barraquer RI, et al. Corneal biomechanics, refraction, and corneal aberrometry in keratoconus: an integrated study. *Invest Ophthalmol Vis Sci*. 2010;51:1948–1955.
9. Roberts CJ, Dupps WJ Jr. Biomechanics of corneal ectasia and biomechanical treatments. *J Cataract Refract Surg*. 2014;40:991–998.
10. Ambrósio R Jr, Correia FF, Lopes B, et al. Corneal biomechanics in ectatic diseases: refractive surgery implications. *Open Ophthalmol J*. 2017; 11:176–193.
11. Moshirfar M, Motlagh MN, Murri MS, et al. Advances in biomechanical parameters for screening of refractive surgery candidates: a review of the literature, Part III. *Med Hypothesis Discov Innov Ophthalmol*. 2019;8:219–240.
12. Vinciguerra R, Ambrósio R Jr, Roberts CJ, et al. Biomechanical characterization of subclinical keratoconus without topographic or tomographic abnormalities. *J Refract Surg*. 2017;33:399–407.
13. Steinberg J, Katz T, Lücke K, et al. Screening for keratoconus with new dynamic biomechanical in vivo scheimpflug analyses. *Cornea*. 2015;34: 1404–1412.
14. Padmanabhan P, Lopes BT, Eliasy A, et al. Evaluation of corneal biomechanical behavior in vivo for healthy and keratoconic eyes using the stress-strain index. *J Cataract Refract Surg*. 2022;48:1162–1167.
15. Francis M, Pahuja N, Shroff R, et al. Waveform analysis of deformation amplitude and deflection amplitude in normal, suspect, and keratoconic eyes. *J Cataract Refract Surg*. 2017;43:1271–1280.
16. Ali NQ, Patel DV, McGhee CNJ. Biomechanical responses of healthy and keratoconic corneas measured using a noncontact scheimpflug-based tonometer. *Invest Ophthalmol Vis Sci*. 2014;55:3651–3659.
17. Alkanaan A, Barsotti R, Kirat O, et al. Collagen fibrils and proteoglycans of peripheral and central stroma of the keratoconus cornea –Ultrastructure and 3D transmission electron tomography. *Sci Rep*. 2019;9:19963.
18. Götzinger E, Pircher M, Dejaco-Ruhswurm I, et al. Imaging of birefringent properties of keratoconus corneas by polarization-sensitive optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2007;48:3551–3558.
19. Maile H, Li JO, Gore D, et al. Machine learning algorithms to detect subclinical keratoconus: systematic review. *JMIR Med Inform*. 2021;9:e27363.
20. Huang W, Randhawa R, Jain P, et al. Development and validation of an artificial intelligence-powered platform for prostate cancer grading and quantification. *JAMA Netw Open*. 2021;4:e2132554.
21. Lee H, Troschel FM, Tajmir S, et al. Pixel-level deep segmentation: artificial intelligence quantifies muscle on computed tomography for body morphometric analysis. *J Digit Imaging*. 2017;30:487–498.
22. Lee A, Taylor P, Kalpathy-Cramer J, et al. Machine learning has arrived!. *Ophthalmology*. 2017;124:1726–1728.
23. Xie Y, Zhao L, Yang X, et al. Screening candidates for refractive surgery with corneal tomographic-based deep learning. *JAMA Ophthalmol*. 2020; 138:519–526.
24. Cao K, Verspoor K, Sahebjada S, et al. Accuracy of machine learning assisted detection of keratoconus: a systematic review and meta-analysis. *J Clin Med*. 2022;11:478.
25. Lin SR, Ladas JG, Bahadur GG, et al. A review of machine learning techniques for keratoconus detection and refractive surgery screening. *Semin Ophthalmol*. 2019;34:317–326.
26. Almeida GC Jr, Guido RC, Balarin Silva HM, et al. New artificial intelligence index based on Scheimpflug corneal tomography to distinguish subclinical keratoconus from healthy corneas. *J Cataract Refract Surg*. 2022;48:1168–1174.
27. Lopes BT, Roberts CJ, Elsheikh A, et al. Repeatability and reproducibility of intraocular pressure and dynamic corneal response parameters assessed by the Corvis ST. *J Ophthalmol*. 2017;2017:8515742.
28. Reddy JC, Rapuano CJ, Cater JR, et al. Comparative evaluation of dual Scheimpflug imaging parameters in keratoconus, early keratoconus, and normal eyes. *J Cataract Refract Surg*. 2014;40:582–592.
29. Ruan J, Xu S, Chen R, et al. EMLI-ICC: an ensemble machine learning-based integration algorithm for metastasis prediction and risk stratification in intrahepatic cholangiocarcinoma. *Brief Bioinform*. 2022;23: bbac450.
30. Salomão MQ, Hofling-Lima AL, Gomes Esporcatte LP, et al. Ectatic diseases. *Exp Eye Res*. 2021;202:108347.
31. Eliasy A, Chen KJ, Vinciguerra R, et al. Determination of corneal biomechanical behavior in-vivo for healthy eyes using CorVis ST tonometry: stress-strain index. *Front Bioeng Biotechnol*. 2019;7:105.
32. Vinciguerra R, Ambrósio R Jr, Elsheikh A, et al. Detection of keratoconus with a new biomechanical index. *J Refract Surg*. 2016;32: 803–810.
33. de Sanctis U, Loiacono C, Richiardi L, et al. Sensitivity and specificity of posterior corneal elevation measured by Pentacam in discriminating keratoconus/subclinical keratoconus. *Ophthalmology*. 2008;115: 1534–1539.
34. Bamdad S, Sedaghat MR, Yasemi M, et al. Sensitivity and specificity of belin ambrosio enhanced ectasia display in early diagnosis of keratoconus. *J Ophthalmol*. 2020;2020:7625659.
35. Lopes BT, Bao F, Wang J, et al. Review of in-vivo characterisation of corneal biomechanics. *Med Novel Technol Devices*. 2021;11:100073.
36. Tan Z, Chen X, Li K, et al. Artificial intelligence-based diagnostic model for detecting keratoconus using videos of corneal force deformation. *Transl Vis Sci Technol*. 2022;11:32.